

Detector-coupled Monte Carlo estimate of background and unresolved-line sensitivity for a balloon-borne 511 keV Laue-lens TES telescope

Authors omitted for review

Abstract

The Galactic 511 keV annihilation signal is dominated by diffuse bulge and disk emission, while compact or point-like contributions remain poorly constrained. We present a detector-coupled Monte Carlo estimate for a balloon-borne pointed telescope combining a 10 m Ge(111) Laue lens with a transition-edge sensor microcalorimeter focal plane. The signal chain transports focused 511 keV photons through the Be aperture into the detector/cryostat mass model. The background chain transports prompt atmospheric particles and photons and production-position-sampled delayed activation decays through the same model. Prompt, delayed, and signal streams are analyzed with a common 510.58–511.42 keV window, CsI anticoincidence veto, and topology/field-of-view consistency selection. For a reference point-source flux of 10^{-4} ph cm $^{-2}$ s $^{-1}$, the selected day-15 signal and background rates are 1.19×10^{-3} s $^{-1}$ and 3.92×10^{-2} s $^{-1}$, respectively; prompt atmospheric events account for 93.4% of the selected background. Folding these rates through the adopted synthetic 20 d trajectory gives $Z_{20d} = 7.80$ and a corresponding 3σ flux threshold of 3.85×10^{-5} ph cm $^{-2}$ s $^{-1}$. The result is a selected-rate flight-sensitivity estimate for a spectrally unresolved compact 511 keV source under the reference geometry and response model. Prompt atmospheric events dominate the selected background, while the delayed component is smaller but materially structured, identifying shielding, side-entry collimation, local materials, and activation validation as the main next engineering and analysis targets.

Keywords Gamma-ray instrumentation, 511 keV annihilation line, transition-edge sensors, Laue lens, balloon-borne telescope, activation background

In this work we develop an end-to-end detector-coupled Monte Carlo chain for this concept and use it to estimate the in-flight background and point-source performance at balloon altitude and to provide directional guidance for design optimization. The chain takes the EXPACS/PARMA atmospheric cosmic-ray field as input and, at balloon altitude, transports the prompt particles while tracking their activation products, yielding both the prompt atmospheric background and the delayed activation background that builds up over the flight; the focused 511 keV reference signal is provided by a 10 m Ge(111) Laue-optics model. The signal and both backgrounds are injected into the same representative balloon-borne TES focal-plane mass model (a dilution-refrigerator cold end, passive shielding, and an active anticoincidence shield), processed with a common narrow energy window, anticoincidence veto, and topology/field-of-view selection, and folded along a reference flight trajectory to give the selected signal rate, the background composition, and the counting significance. The selected background is dominated by the prompt atmospheric component, so the shield geometry, side-entry collimation, and shield material are the most direct directions for design optimization; the delayed activation and the upstream optics-hardware background enter the same evaluation as secondary components. This work thereby estimates the in-flight background and detection performance of the balloon platform and indicates directions for design optimization.

0.1 Science and instrument requirements

The simulated instrument is aimed at 511 keV point-source or unresolved-central-excess components from the Galactic-center region that have not been detected by previous instruments. Existing INTEGRAL/SPI measurements have established a 511 keV sky dominated by diffuse bulge and disk emission and limit point-like 511 keV sources superposed on that diffuse emission to fluxes of order 10^{-4} ph cm $^{-2}$ s $^{-1}$; an independent INTEGRAL/IBIS compact-source search also found no Galactic 511 keV point sources and gives an upper limit of about 1.6×10^{-4} ph cm $^{-2}$ s $^{-1}$ for a ~ 10 Ms Galactic-center exposure [Knoedlseder et al.(2005), De Cesare(2011)]. For a representative 511 keV atmospheric transmission of $T \simeq 0.739$ at balloon altitude, this limit corresponds to an above-payload photon-flux scale of about 1.2×10^{-4} ph cm $^{-2}$ s $^{-1}$. The simulation therefore adopts $F_0 = 10^{-4}$ ph cm $^{-2}$ s $^{-1}$ as the reference point-source photon-flux scale. Compared

with the wide-field spectroscopic and coded-mask searches performed by INTEGRAL, the modeled payload adds Laue focusing to improve the point-source response and uses a high-energy-resolution, multilayer thick-absorber TES microcalorimeter array as the focal-plane detector to sharpen the 511 keV line measurement [Shirazi et al.(2023), Caputo et al.(2017)].

This work first considers a balloon payload as an early test platform, with the expectation that the approach can later be extended to a satellite observing platform.

This goal imposes four requirements on the instrument and the simulation: first, the telescope must concentrate 511 keV photons onto a small focal plane, so that detector-associated background is reduced without sacrificing collecting area. Second, the focal-plane response must support a sub-keV energy window, so that the line-window background scales with a narrow spectral interval rather than a broad MeV-band continuum. Third, the detector and shield model must suppress atmospheric, activation, and side-entry backgrounds while explicitly accounting for the passive material near the TES absorbers. Fourth, the simulation chain must treat prompt background, delayed activation, focused signal, vetoes, event topology, and mission-time variation consistently, so that signal and background are evaluated under the same detector selection.

The remainder of this paper is organized as follows. Section 1 describes the detector and cryostat mass model; Section 2 the simulation framework—the workflow, optics coupling, background sources, event selection, and mission-time fold; Section 3 the selected-rate and sensitivity results together with the normalization, convergence, and boundary checks; and Section 4 the discussion and conclusions.

1 Detector and cryostat mass model

The detector mass model is an engineering Geant4/MEGAlib proxy mass model that contains a balloon-borne multilayer TES array mounted at the cold end of a compact dilution refrigerator (DR) [Shirazi et al.(2023), Gottardi & Nagayoshi(2021)]. The detector model also contains typical DR cryostat structures, active shielding, passive shielding, magnetic shielding, side-entry windows, collimation structures, copper thermal structures, and other cryogenic service-mass proxies. Figure 1 shows two-dimensional views of the mass model used in this paper.

The TES detector concept uses an absorber-coupled AlMn film structure developed in our laboratory; the advantages of AlMn alloy include transition-temperature tunability through composition and annealing, suitability for array fabrication, and relatively low magnetic-field sensitivity [Xie et al.(2026), Vavagiakis et al.(2017)]. In the present concept, the TES thermometer film is an AlMn film of order 200 nm thickness, whereas the γ -ray absorber is a millimetre-scale metal body. The absorber uses $1.5 \times 1.5 \times 3.0 \text{ mm}^3$ Ta pixels; the high atomic number of Ta increases the interaction probability for 511 keV photons, and a 3.0 mm thickness corresponds to about a 49% single-layer interaction probability at 511 keV, so the six-layer array can provide a near-unity geometric interaction probability. In the current geometry, the center-to-center spacing between adjacent Ta array layers is 12 mm; this spacing is used to preserve the multilayer hit topology required for the Compton-kinematic veto discussed below. The choice of Ta also considers that this volume can provide a sufficiently high saturation energy, low low-temperature specific heat (this paper uses an order-of-magnitude estimate of $10^{-2} \text{ J K}^{-1} \text{ m}^{-3}$), and a relatively high photoelectric-to-Compton interaction ratio; based on NIST XCOM cross sections near 511 keV, this ratio is about 0.731 [Hubbell & Seltzer]. Based on laboratory experience and by substituting the current Ta absorber heat capacity of 67.5 pJ/K, the simulation adopts $\Delta E_{\text{FWHM}} \simeq 420 \text{ eV}$ at 511 keV as the current simulated energy resolution.

Because the TES film and Ta absorber differ greatly in size and mass, and because the TES film itself does not contribute significant 511 keV interactions, the transport geometry uses the Ta absorber to represent the main sensitive volume of a TES pixel. The explicit sensitive volume has six layers, 376 Ta absorber pixels per layer, and 2256 pixels in total, with 1.5 mm length and width and 3 mm thickness. A Si substrate proxy is placed below each Ta pixel, using Si to represent the substrate; this also preserves the elastic-scattering contribution that silicon may provide in neutron transport. TES-film, wiring, and readout details are not built layer by layer as explicit interaction volumes; instead, they enter the model through the assumed energy response and service-mass proxies. In the current manuscript, the TES energy resolution is used as the 511 keV line-window width, with $W_{511} = 511 \text{ keV} \pm 420 \text{ eV}$, i.e. 510.58–511.42 keV.

The TES array is mounted on the required copper support and thermal-link structures. Copper provides the decisive thermal-conduction path; a copper thermal finger exits through the bottom hole of the sample box and connects to the 50 mK cold plate. The near-field TES sample box serves as both magnetic shielding and local passive shielding. The current mainline geometry consists of a 2 mm Nb inner layer and a 2 mm MuMetal outer layer, with MuMetal represented in the material table as Ni:Fe = 4:1. The sample-box entrance window is covered by a thin Al window; the rear end cap retains the hole for the cold finger. A W–Nb double-layer sample box is treated as a comparison passive-shielding + magnetic-shielding material design; the X-ray window is opened on the side of the cryostat, with a 300 μm thick Al window embedded in the thermal-shield shell inside each cryostat layer, a 25 μm thick Be window placed in the vacuum shell, and an x -thick W collimator placed outside the Be window, with the collimator opening strictly aligned to the TES pixels. The side window is strictly defined through Boolean operations. The mass model also introduces scintillator active shielding, wrapped around the outer side wall, bottom, and top of the cryostat. Segmented CsI active shielding is used, with a radial side-shield thickness of 4 cm; the bottom and top shields are included in the geometry as axial end volumes. CsI volumes enter the simulation as transport and activation materials, and their deposited energies are read in post-processing to apply the active veto; the counting threshold is 50 keV. BGO active shielding is treated as a material-comparison case; the BGO threshold is 70 keV. Considering the physical layout of the DR and the side-entry optical path, the cryostat instrument frame is rotated as a whole by 45 degrees, so that the local side window is tilted 45° upward in the zenith-frame source coordinates.

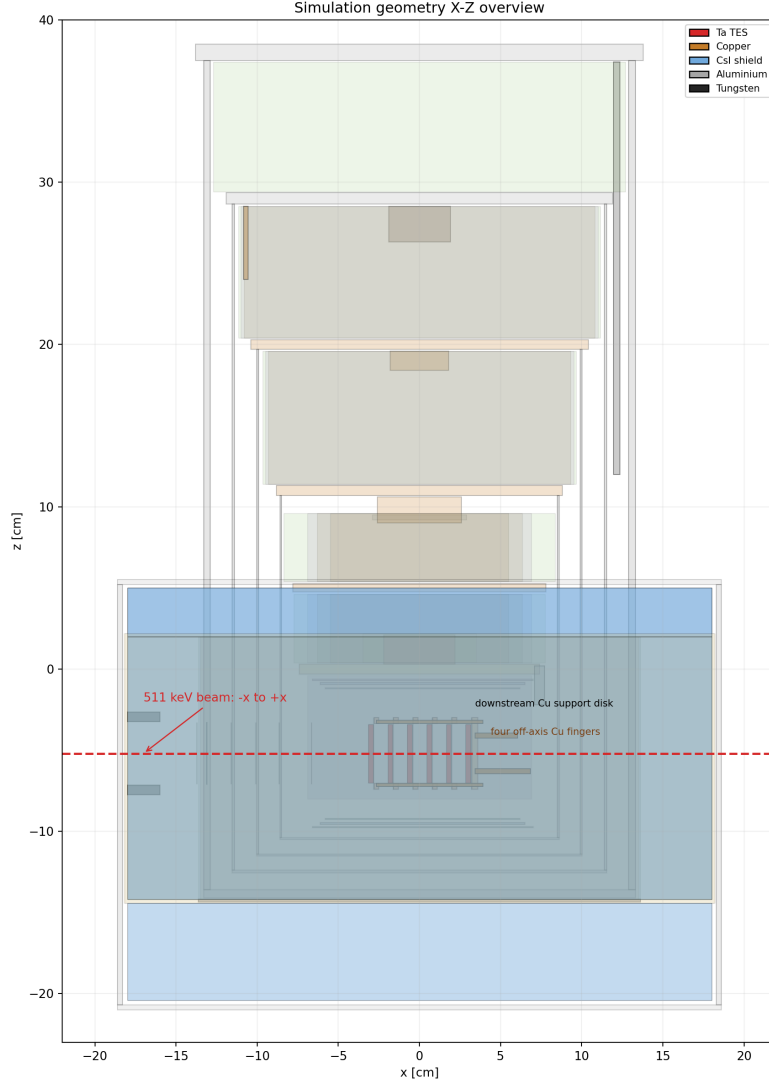


Figure 1: Two-dimensional overview of the detector/cryostat proxy mass model, showing the X-Z view of the DR tower, detector bay, active shield, and 511 keV side-entry beam path.

2 Simulation framework

This section describes the whole simulation chain from source construction to flight-performance assessment. The focused signal and detector background start from two mass models. The focused-signal branch is implemented in the Laue-optics model 10m upstream, focusing 511 keV photons onto the focal plane, **also considering the influence of the instantaneous background and activation background on this mass model**, and outputting the focal-plane slice together as the focused signal. The detector-background branch directly simulates, in the detector/cryostat mass model of Section 2, the delayed background caused by prompt atmospheric cosmic rays and activated nuclides. The prompt atmospheric cosmic rays all use the far-field definition defined by megalib, according to the zenith-angle flux output by the expacs library, with full-solid-angle input. For activated nuclides, the real positions where nuclides are produced are recorded, and then activation background is generated by sampling according to the production rate. The focused signal, instantaneous background, and activation background are normalized and superposed to build the total spectrum. After detector response, anticoincidence, and event selection based on Compton tracks inverted from different pixels, the effective spectrum is used for post-processing statistical analysis and flight-performance assessment.

The above simulation is for a specific balloon-flight moment, longitude and latitude, and altitude. Although for an observation time such as one hour equivalent to several Monte Carlo events, this position can be regarded as constant. However, for a long-time flight on the scale of days, the altitude, longitude, and latitude obviously

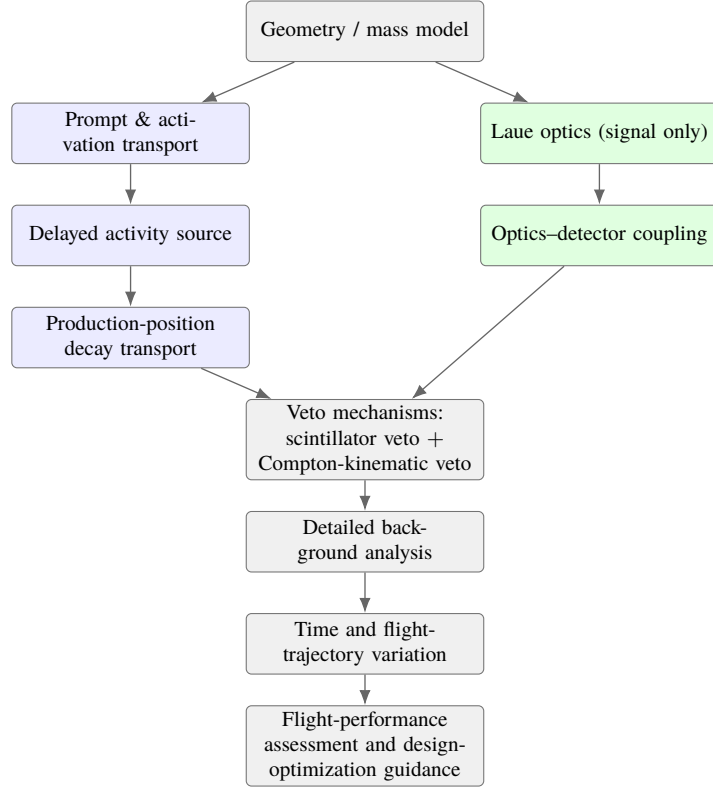


Figure 2: Simulation chain used in this work. Blue nodes are the prompt and delayed background streams, green nodes the focused-signal optics, and grey nodes the shared geometry, veto mechanisms, detailed background analysis, time/trajectory variation, and final flight-performance assessment. The focused signal and all background streams are evaluated under the same detector response and veto definitions before background decomposition, mission-time variation, and design-optimization judgments are made.

change significantly. This causes changes in the magnetic-rigidity cutoff for instantaneous cosmic rays and changes in the atmospheric transmission for the signal, and the activation background itself also changes with the incident cosmic-ray flux and different stages of the decay chain. This reminds us that the background-signal at a reference moment should not be linearly extrapolated in time to evaluate the time-folded flight performance over the whole working time. Therefore, we introduce a reasonable analytical flight-trajectory–time curve to evaluate the performance over the whole working cycle, and simulate multiple points on the curve to verify the rationality of the curve.

2.1 Laue optics and detector-coupled signal replay

This simulation adopts a laue-lens system based on mosaic Ge(111) crystals [Barriere et al.(2009), Barriere et al.(2011)]. Through optical design, it realizes a 10 m focal length, **a multi-ring system realizes the 450keV-550keV detection band**, and an effective area of 20cm**2 at 511keV; the effective area is calculated as

$$A_{\text{eff}}(E) = A_{\text{inc}} \frac{N_{\text{fp}}(E)}{N_{\text{inc}}(E)} \quad (1)$$

where A_{inc} is the Monte Carlo incident sampling area, $N_{\text{inc}}(E)$ is the number of incident photons, and $N_{\text{fp}}(E)$ is the number of photons satisfying the focusing/focal-plane interception condition. The mosaic system is adopted to make the Monte Carlo calculation convenient while satisfying the expected focusing performance; other candidate schemes are also possible for actual engineering adaptation.

In this simulation, the laue optical branch gives the focusing-performance simulation and **also realizes particle transport of itself as mass**.

For this purpose, the diffraction probability of photons by the crystal is no longer superposed onto photons as an efficiency parameter, but is defined as a discrete process at the Geant4 application layer

[Agostinelli et al.(2003), Allison et al.(2016)]. The process takes the diffraction probability $R(|\Delta\theta|)$ according to the angular deviation of the current photon relative to the crystal plane, and converts it into a finite equivalent diffraction mean free path λ_{diff} , so that the accumulated diffraction probability when the photon crosses a crystal path ℓ reproduces this value: $1 - \exp(-\ell/\lambda_{\text{diff}}) = R(|\Delta\theta|)$, namely $\lambda_{\text{diff}} = -\ell/\ln[1 - R(|\Delta\theta|)]$; to avoid double counting with standard photoelectric absorption, R is first normalized by the unabsorbed fraction ($R/(1 - A)$) and then converted into a mean free path. After this mean free path is returned to Geant4, the diffraction process and the standard electromagnetic processes (photoelectric absorption and Compton scattering) each sample interaction lengths in the same competition framework, and the shortest one decides which process occurs first, rather than forcibly intercepting photons at the crystal boundary. Photons not selected to undergo diffraction still continue transport according to the standard electromagnetic physics, and may be absorbed, scattered, or directly transmitted.

The diffraction probability is preferentially given by interpolation in an external XOP/CRYSTAL rocking-curve map as a function of angular deviation: this curve is calculated offline once in the preprocessing stage by the CRYSTAL/diff_pat module of the XOP toolkit (called through xoppylib, with DABAX crystallographic data as input) [Sanchez del Rio & Dejus(2011)], and gives the Ge(111) reflectivity/transmissivity/absorption curves as a function of angular deviation; during transport, the angular deviation is calculated in real time for every step inside the crystal, and the current diffraction probability is obtained by interpolation on that curve, namely “offline library tabulation, probability used by angle during transport”. The mosaic orientation is implemented by applying a Gaussian angular perturbation to the ideal Bragg-plane normal, with the perturbation width given by the 30 arcsec mosaic FWHM; after diffraction occurs, the outgoing direction is mirror-reflected according to the perturbed plane normal, and the photon energy is unchanged. This split in which the “process class only handles the transport interface, and the independent model class handles probability and direction decisions” follows the idea of existing crystal Bragg-reflection extensions in Geant4 [Guan et al.(2023), Reiazi et al.(2025)], and extends it to the energy band needed here.

Here the signal photons use as reference the point source of the V404 source once detected by SPI [Siegert et al.(2016)]. V404 is a transient point source in the Galactic disk and is used here only as a reference. It is constructed using the megalib far-field source mode [Zoglauer et al.(2006)]; **the detailed construction method is given in Section 3.2.3 on point-source construction.** In addition to signal introduction, we also introduce the prompt cosmic-ray background component, also constructed in the far-field source mode, and introduce the activation-background construction. **The transport of the above three sources is simulated separately; in the three simulations, the particle flux, species, and energy crossing the plane where the focal plane is located are all intercepted as outputs; the three simulation outputs are normalized by equivalent observation time as the total output, and this total output can be separately input into the detector mass model as a parallel-beam source.**

2.1.1 Reference point-source construction

The focused signal stream is defined by a reference astrophysical point source. As a concrete example of the target class we take the 511 keV emission episode of the microquasar V404 Cygni observed with INTEGRAL/SPI [Siegert et al.(2016)]; V404 Cygni is a transient point source in the Galactic disk and serves here only as a reference for the source class, not as a predicted target flux. The reference flux scale itself, $F_0 = 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$, is the above-atmosphere value motivated in Section 0.1.

In the optics run, the source is a monochromatic 511 keV far-field beam: photons arrive parallel to the optical axis, as from a point source at astronomical distance, and are sampled uniformly over the incident aperture A_{inc} of Eq. (1). Three independent seeds of 5×10^4 primaries each give 1.5×10^5 transported photons, of which 37,256 are diffracted and cross the focal plane; 37,194 of these (99.8%) fall within the $r = 1.898 \text{ cm}$ Be entrance window, with focused-spot radii $r_{90} = 1.03 \text{ cm}$ and $r_{99} = 1.25 \text{ cm}$. The within-window crossings form the event list that is replayed through the Be aperture into the detector/cryostat model with the recorded positions and directions.

The normalization ties this replay to the reference flux. At the detector entrance the focused stream carries the rate $F_0 A_{\text{eff}}(511 \text{ keV}) T$, where $T = 0.739$ is the reference 511 keV atmospheric transmission at the day-15 trajectory anchor (Section ??); numerically this is $1.48 \times 10^{-3} \text{ s}^{-1}$ at the reference flux. Equivalently, the

generating run corresponds to an observation time $N_{\text{fp}}/[F_0 A_{\text{eff}} T]$, and each replayed event carries the inverse of that exposure as its rate weight; in the intermediate cut-flow bookkeeping of Section ?? the same sample appears as a unit replay before this scale is applied. After the common detector response and event selection, the surviving signal rate is $1.19 \times 10^{-3} \text{ s}^{-1}$, corresponding to a selected effective area of $S_{W_{511}}/F_0 = 11.9 \text{ cm}^2$. Because the optics run contains only source photons, the replayed stream is signal-pure; all backgrounds enter through the streams of Section 2.2 and the boundary terms of Section ??.

2.2 Prompt atmospheric and delayed-activation source model

With the focused signal defined in Section 2.1.1, the background side of the calculation requires two further source layers: the prompt atmospheric field incident on the payload, and the delayed radioactive population that this field builds up in the instrument. The two are constructed separately, so that source generation does not absorb assumptions that belong to detector selection: everything in this section defines incident particles or decaying nuclei, while vetoes, topology cuts, and the hard line window are applied only in Section ??.

2.2.1 Prompt atmospheric source

The prompt atmospheric field is derived from EXPACS/PARMA. PARMA3.0 is an analytical model parameterized from PHITS atmospheric-shower calculations and benchmarked against terrestrial cosmic-ray measurements, and EXPACS is its public tabulation and software interface [Sato(2015), Sato(2016)]. PHITS supplies the underlying particle-transport physics: primary cosmic rays interact with air nuclei, the secondary hadrons, leptons, photons, and neutrons are transported through the atmospheric column, and the resulting fluences are parameterized by altitude, geomagnetic cutoff, solar modulation, energy, and direction. In this work EXPACS/PARMA provides the differential incident flux as a function of particle species, kinetic energy, zenith angle, geographic position, altitude, cutoff rigidity, and solar activity, and is used only to construct the external radiation fields; detector response, vetoing, topology selection, and mission-time counting are applied downstream. The reference source definitions use latitude 34° , longitude 100° , altitude 38 km, vertical cutoff rigidity $R_c = 11.6 \text{ GV}$, and the 2025-08-31 solar condition corresponding to $W = 118.3$.

The prompt source includes photons, neutrons, electrons, positrons, protons, alpha particles, and negative and positive muons. For each particle family the EXPACS/PARMA differential spectrum is converted into a MEGAlib far-field area source covering the full zenith range $\theta = 0^\circ\text{--}180^\circ$ and full azimuth range $\phi = 0^\circ\text{--}360^\circ$. The zenith dependence is represented by 20 differential equal- μ angular bins, where $\mu = \cos \theta$ and each bin subtends $\Delta\Omega = 0.6283185307 \text{ sr}$: the first ten bins cover down-going particles, the last ten up-going or albedo-like particles. Both hemispheres are retained in the prompt transport and in the activation-production transport, so upward albedo particles are not discarded at source construction. Figure 3 shows the resulting full-sphere flux field.

The integrated source-plane fluxes are approximately 4.80, 0.462, 0.199, 0.117, 0.112, 0.0115, 0.0050, and $0.0056 \text{ cm}^{-2} \text{ s}^{-1}$ for γ , n , e^- , e^+ , p , α , μ^- , and μ^+ , respectively; the non-photon families together carry about $0.91 \text{ cm}^{-2} \text{ s}^{-1}$, or 16% of the total. If Monte Carlo counts were allocated strictly in proportion to flux, the detector and activation statistics of these minority families would be poor, so the non-photon families are generated with replicated samples and de-weighted in the rate normalization. All particles are then transported through the detector/cryostat mass model with Geant4/MEGAlib [Agostinelli et al.(2003), Allison et al.(2016), Zoglauer et al.(2006)]; the prompt run and the activation-production run each generated 25,210,216 incident primaries. Each particle species is normalized with its own transported exposure,

$$r_{\text{event},j} = \left(\sum_i \mathcal{T}_{ij} \right)^{-1}, \quad (2)$$

where $\mathcal{T}_{ij}$ is the transported exposure of subsample i for species j . This per-species normalization is required because the photon and non-photon streams have different sampling and exposure structures; the normalization check enforces $r_{\text{event},j} \sum_i \mathcal{T}_{ij} = 1$ for every species, so over-sampling changes the Monte Carlo variance but not the expected physical rate.

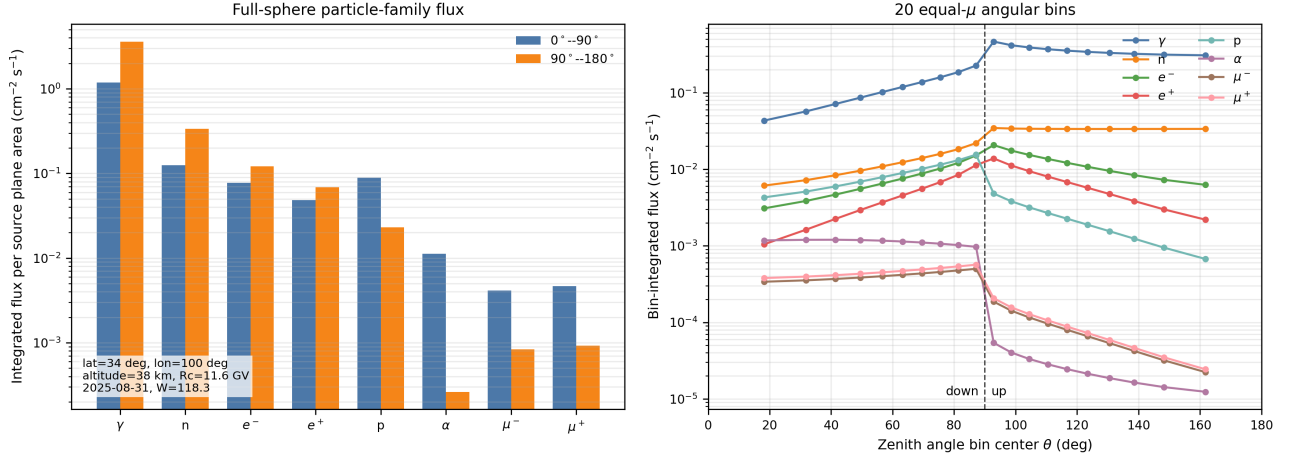


Figure 3: EXPACS/PARMA full-sphere atmospheric flux field used to generate the prompt and activation-production transport inputs. Left: energy- and solid-angle-integrated fluxes for the eight transported particle families, separated into down-going and up-going hemispheres. Right: bin-integrated flux in the 20 equal- μ differential zenith bins used by the MEGALib far-field source. The plot is generated from the source-definition inputs for latitude 34°, longitude 100°, altitude 38 km, $R_c = 11.6$ GV, and $W = 118.3$.

2.2.2 Delayed activation source

The activation-production stream is not counted as an immediate detector background. It records instead the isotope, the material volume, and the production position of every radioactive product. These records are converted into a fixed day-15 radioactive population using production–decay buildup, with ground-state half-lives checked against NUBASE2020 [Kondev et al.(2021)]. For isotope k ,

$$A_k(t_{\text{flight}}) = P_k [1 - \exp(-\lambda_k t_{\text{flight}})], \quad \lambda_k = \frac{\ln 2}{t_{1/2,k}}, \quad (3)$$

where P_k is the production rate inferred from the activation-production transport and $t_{\text{flight}} = 15$ d for the reference state. The resulting fixed delayed activity used by the reference construction is 85.45 Bq.

The position information does not come from the aggregate isotope store alone. The standard store records volume names and isotope counts; the production-position construction adds a custom hook in the Geant4/MEGALib stepping-action layer. At the moment an isotope is stored during activation buildup, the hook writes a production-position record containing the logical volume, the pre-step position (x, y, z) , the isotope identifier $1000Z + A$, the excitation energy, time, process, and track-ancestry metadata. The fixed day-15 population is then joined to these records by volume, isotope, and excitation state, which gives the delayed source access to the simulated production coordinates rather than only to volume-integrated activities. Figure ?? in Appendix ?? shows the resulting position distribution: the activity-weighted map is dominated by CsI-shield activation and ¹²⁸I, with smaller contributions from Cu, Al, W, Mg, and Cs isotopes. It is a source-construction diagnostic, not a detector-count map; detector response is applied only after the delayed decays are transported through the common geometry and event selection.

Whether the production positions are worth preserving can be tested from the same table. Delayed decays are emitted locally, so downstream attenuation, active-shield veto probability, and event topology all depend on where the isotope was produced, and different incident families—with their different energy-loss, stopping, and capture histories—need not deposit their activation in the same materials. To quantify this, the weighted production rows are grouped by incident particle family and material category, and the activity-normalized category distributions of two families a and b are compared with the total-variation distance

$$D_{\text{TV}}(a, b) = \frac{1}{2} \sum_c |f_{a,c} - f_{b,c}|, \quad (4)$$

where $f_{a,c}$ is the fraction of delayed activity from family a produced in material category c . In the fixed day-15 table, neutron-induced products carry about 82.5 Bq, μ^- -induced products 2.75 Bq, proton-induced products

0.15 Bq, and α -induced products 0.022 Bq, with smaller μ^+ and e^+ terms. The neutron-induced activity remains dominated by the CsI shield, whereas the μ^- component is qualitatively different: Cu/support material, Al shells, and W-bearing passive material account for 32%, 29%, and 21% of the μ^- -induced activity, respectively (Figure ?? in Appendix ??). A volume-integrated or axisymmetric radial source could preserve the total activity, but it would erase these incident-family, material, nuclide, and coordinate correlations; the production-position construction keeps them in the source before detector transport.

for the baseline CsI configuration. The shield sum is evaluated from the CsI active-shield volumes in post-processing. Native detector trigger/veto blocks are present in the geometry description, but the quantitative rate definition is this common post-processing definition, so prompt, delayed, and focused-signal streams use the same coincidence window and threshold.

The second layer is a topology/field-of-view consistency test on the TES hits that survive the active-shield gate. Baseline event classes are kept explicit: single-site events, valid reconstructed multi-site events, and unreconstructed multi-site events. Single-pixel TES events are retained as calorimetric line candidates. For a two-hit event, both possible first-scatter orders are tested under the assumptions of a single Compton scatter followed by full absorption in the measured TES hits. For an assumed order with first deposited energy E_1 and remaining scattered-photon energy E_{sc} , the kinematic angle is computed from

$$\cos \theta_C = 1 - m_e c^2 \left(\frac{1}{E_{sc}} - \frac{1}{E_1 + E_{sc}} \right). \quad (5)$$

The event is kept if either order gives a physically valid first-scatter cone that intersects the side Be-window disk. For higher multiplicities the current implementation enumerates candidate hit orders up to six TES hits, selects the best sequence using the unweighted Compton sequence residual

$$Q = \sum_{i=1}^{n-2} (\cos \theta_{C,i} - \hat{\mathbf{u}}_{i-1,i} \cdot \hat{\mathbf{u}}_{i,i+1})^2, \quad (6)$$

and then applies the same first-cone intersection test to the first reconstructed scatter. If no valid sequence is found, the event is classified as unreconstructed. Such events are retained in the baseline selected-rate definition but are not called field-of-view passing. The line-window result therefore does not gain sensitivity from rejecting ambiguous reconstruction failures. This policy is not a conservative lower bound on the net sensitivity: in the current W_{511} sample, a single-site-only diagnostic alternative that vetoes all multi-site candidates would retain 59.4% of the focused signal and 68.2% of the selected background, reducing the simple $S/\sqrt{B}$ figure to 0.720 of the baseline value. The baseline selection therefore treats ambiguous multi-hit events as a reconstruction-model boundary that must be validated, not as a sensitivity gain from aggressive event rejection.

To connect this selection chain explicitly to the finite Monte Carlo sample, Table 1 gives the current event-level W_{511} cut-flow obtained by lightweight replay of the simulated event catalogues. The table is not a new Geant4/MEGAlib transport. It replays the current event catalogues with the same TES energy window, CsI veto, and Compton-kinematic/field-of-view consistency checks; uncertainties are Monte Carlo quadrature sums over event weights. The current event-selection records first fix the energy window and then report raw, active-shield, and topology stages, so the raw stage should be read as the pre-veto sample inside the line window, not as an all-energy incident sample.

The detector-level time-axis construction used to validate this veto bookkeeping is a Poisson superposition of independent streams. For stream k with event-catalogue rates r_{kj} and total rate $R_k = \sum_j r_{kj}$, a validation 81 points from 0 to 20 d with 0.25 d spacing; endpoint points carry half-widths in the count integration. The trajectory is a synthetic reference profile, not balloon telemetry. For time-grid point t_i in days,

$$h_i = h_0 + \Delta h \sin \left[\frac{2\pi(t_i - 15)}{5} \right], \quad (7)$$

$$\varphi_i = \varphi_0 + \Delta \varphi \sin \left[\frac{2\pi(t_i - 15)}{7} \right], \quad (8)$$

$$\ell_i = \ell_0 + \Delta \ell \sin \left[\frac{2\pi(t_i - 15)}{9} \right], \quad (9)$$

Table 1: event-level W_{511} line-window cut-flow.

Stream	Stage	Events	Rate / s^{-1}	Statistical uncertainty / s^{-1}
Prompt background	Line-window pre-veto	161	1.1877×10^{-1}	1.15×10^{-2}
Prompt background	After CsI veto	60	4.0712×10^{-2}	5.26×10^{-3}
Prompt background	Topology/FOV selected	54	3.6641×10^{-2}	4.99×10^{-3}
Delayed activation	Line-window pre-veto	54	4.6354×10^{-3}	6.31×10^{-4}
Delayed activation	After CsI veto	33	2.8327×10^{-3}	4.93×10^{-4}
Delayed activation	Topology/FOV selected	30	2.5752×10^{-3}	4.70×10^{-4}
Focused signal	Line-window pre-veto	30323	8.1527×10^{-1}	4.68×10^{-3}
Focused signal	After CsI veto	30323	8.1527×10^{-1}	4.68×10^{-3}
Focused signal	Topology/FOV selected	29715	7.9892×10^{-1}	4.63×10^{-3}

The focused-signal rows are unit-replay rates used to show veto survival behavior; the point-source signal rate in Table 2 is scaled separately by the Laue effective area, reference flux, and atmospheric transmission.

where $h_0 = 38.75$ km, $\Delta h = 2.5$ km, $\varphi_0 = 34^\circ$, $\Delta\varphi = 0.25^\circ$, $\ell_0 = 100^\circ$, and $\Delta\ell = 0.25^\circ$. The resulting ranges are 36.25–41.25 km in altitude, 33.75° – 34.25° in latitude, and 99.75° – 100.25° in longitude.

The atmospheric depth and the 511 keV transmission scale are

$$X(h_i) = X_{38} \exp\left[-\frac{h_i - 38 \text{ km}}{H}\right], \quad X_{38} = 3.87 \text{ g cm}^{-2}, \quad H = 6.8 \text{ km}, \quad (10)$$

$$T_i = \exp[-\mu_{\text{eff}} X(h_i)], \quad \mu_{\text{eff}} = -\frac{\ln T_{15}}{X(h_0)}, \quad (11)$$

with $T_{15} = 0.739$. Thus the focused-science scale is $f_{\text{atm},i} = T_i/T_{15}$. The prompt atmospheric source scale is

$$R_{c,i} = 11.0 - 0.08(\varphi_i - \varphi_0) + 0.03(\ell_i - \ell_0), \quad (12)$$

$$f_{p,i} = \exp\left[\frac{X(h_i) - X(h_0)}{30 \text{ g cm}^{-2}}\right] \left(\frac{11.0}{R_{c,i}}\right)^{0.2}. \quad (13)$$

The trajectory rigidity model is centered on a rounded value, and $f_{p,i}$ enters only through the day-15–anchored ratio $(11.0/R_{c,i})^{0.2}$, so this central value affects the mission-time variation but not the absolute prompt rate, which is fixed by the $R_c = 11.6$ GV source definition. The same scalar production driver is used for delayed activation before isotope-by-isotope decay buildup is solved. For a delayed-production element a with decay constant λ_a and fixed day-15 activity $A_{a,15}$,

$$P_{a,15} = \frac{A_{a,15}}{1 - \exp(-\lambda_a t_{15})}, \quad (14)$$

$$\frac{dN_a}{dt} = P_{a,15} f_p(t) - \lambda_a N_a, \quad A_a(t) = \lambda_a N_a(t). \quad (15)$$

The numerical fold anchors the solution back to $A_{a,15}$ at day 15 and sums the activity over the nuclide/material/position production elements.

This delayed-activity treatment assumes that the normalized activation-product distribution is unchanged over the small trajectory range, and that the trajectory changes only the scalar production driver. The verification described here checks the internal implementation of that model but does not prove physical invariance of the production distribution. If dedicated activation transports at the trajectory center and extrema show material or nuclide redistribution, the scalar driver should be replaced by interpolation over production vectors,

$$P_a(t_i) = \sum_g w_g(h_i, \varphi_i, \ell_i, R_{c,i}) P_a^{(g)}, \quad \sum_g w_g = 1, \quad (16)$$

followed by the same decay ODE.

Implementation checks recomputed the trajectory formula at the center point and at the two altitude extrema. The reference atmospheric transmission is $T = 0.739$ at the day-15 anchor, and the high- and low-altitude probes

give $T = 0.811$ and 0.646 , respectively. Reconstructing the W_{511} prompt, delayed, background, and signal rates from the day-15 rates and these scales reproduces the mission-fold summary. Reweighting the particle families with local PARMA spectra at the same trajectory probes shifts the normalized nuclide–volume activity distribution by less than 6×10^{-3} in total-variation distance. This supports using a scalar delayed production/activity fold for the present small trajectory range at the particle-family reweighting layer. It is not a full proof of distribution invariance: same-particle energy/angle-dependent activation cross sections and position-transport shifts would require dedicated low-, reference-, and high-altitude activation-production transports.

In the mission-time fold, the selected day-15 rates are folded over the trajectory grid as

$$B_i = L_i (B_{p,15} f_{p,i} + B_{d,15} f_{d,i}), \quad (17)$$

$$S_i = L_i S_{15} f_{\text{atm},i}, \quad (18)$$

$$L_i = \exp[-(R_{p,i} + R_{d,i})\tau], \quad (19)$$

where $f_{p,i}$ is the prompt atmospheric scale, $f_{d,i}$ is the delayed-activity scale, $f_{\text{atm},i}$ is the 511 keV atmospheric transmission scale, and L_i is the accidental coincidence live factor. Here $R_{p,i}$ and $R_{d,i}$ are the prompt and delayed event rates entering the coincidence grouping at that mission point. The factor L_i assumes a one-sided τ gate; a symmetric $\pm\tau$ condition would instead give $\exp[-2(R_{p,i} + R_{d,i})\tau]$, but for the present rates both forms remain very close to unity, so the folded counts are insensitive to this choice.

The time-dependent significance is computed as a cumulative counting statistic. For a reference flux F_0 , the cumulative significance is

$$Z(t_m) = \frac{N_s(t_m)}{\sqrt{N_b(t_m)}}, \quad (20)$$

where

$$N_s(t_m) = \sum_{i \leq m} S_i \Delta t_i, \quad N_b(t_m) = \sum_{i \leq m} B_i \Delta t_i \quad (21)$$

are the time-integrated selected signal and background counts after mission-time scaling and live-factor corrections. For comparison among analysis variants, the 3σ flux threshold is scaled as

$$F_{3\sigma}(t_m) = F_0 \frac{3}{Z(t_m)}. \quad (22)$$

This Gaussian counting convention is used consistently for the line-window

3 Results

The recovered log fragments around the original detailed results tables are structurally mixed. This compile-ready copy therefore keeps the confirmed primary selected-rate numbers in a compact recovery table and leaves the original detailed table reconstruction to manual editing.

Table 2: Recovered primary unresolved-line selected-rate result.

Quantity	Value	Note
Day-15 selected background	$3.92 \times 10^{-2} \text{ s}^{-1}$	Prompt + delayed
Day-15 selected signal at F_0	$1.19 \times 10^{-3} \text{ s}^{-1}$	$F_0 = 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$
Prompt background fraction	93.4%	Dominant component
20 d diagnostic Z	7.80	Counting metric
$F_{3\sigma}(20 \text{ d})$	$3.85 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$	Reference model only

Recovery gap note

The original recovered log fragments become structurally inconsistent after this point. The broad-window diagnostic paragraph and several nearby transition lines are omitted from this compile-ready reading copy; see the raw partial source and recovery manifest.

4 Discussion and conclusions

The main result is that the reference production-position delayed-source calculation gives a 20 d diagnostic 3σ point-source threshold of $3.85 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$ for an unresolved line in the modeled focused TES line spectrometer. This threshold is obtained without using the focused-spot spatial analysis as the primary result. It follows from three combined effects: optical concentration, a sub-keV energy window, and a detector-coupled veto/topology selection that suppresses prompt and delayed backgrounds in the same selection space as the signal.

This calculation should be compared with existing 511 keV instruments as a pointed, narrow-line, compact-source estimate, not as an all-sky imaging forecast. INTEGRAL/SPI and COSI provide the reference measurements for mapping the diffuse 511 keV sky and the broader Galactic diffuse emission [Siegert et al.(2016), Kierans et al.(2020), Siegert et al.(2020), Yoneda et al.(2025), Karwin et al.(2023)]. Peer-reviewed COSI calibration and simulation studies provide the more relevant technical comparison for detector response, event reconstruction, and balloon-background validation [Beechert et al.(2022), Sleator et al.(2019), Gallego et al.(2025a), Gallego et al.(2025b), Ciabatonni et al.(2025)]. The TES Laue-lens concept considered here is complementary: it trades sky coverage for a concentrated beam and a sub-keV line window. The relevant comparison class is compact 511 keV targets, unresolved-central-excess searches, and transient follow-up observations with known or constrained source positions. The same focused aperture is not a substitute for measuring the full diffuse bulge and disk luminosity, because it samples only a small solid angle.

The production-position delayed-source treatment addresses a specific approximation in radial-profile compression. Although delayed activation contributes only about 6.6% of the selected line-window background in the present calculation, it is spatially and materially structured. Preserving sampled production positions prevents artificial azimuthal or radial redistribution from becoming an unquantified systematic. The delayed component has passed a minimum selected-rate convergence check using four independent production-position samplings, but this should not be read as a complete delayed-background systematic envelope.

The background composition indicates the first engineering parameters to test, rather than a completed optimization. The selected line-window rate remains dominated by prompt atmospheric streams, so shield geometry, side-entry collimation, material near the focal plane, topology selection, and multi-pixel reconstruction are the most direct handles on the present background. Delayed activation is smaller but remains spatially and materially structured, so source-position preservation is retained in the rate calculation. The dominance of iodine activation in the CsI shield also motivates a future shield-material trade study.

4.1 Limitations and future validation

The selected-rate result has five immediate boundaries. First, the detector/cryostat model is an engineering proxy, not a final flight CAD model. Second, quantitative active-shield rejection is applied in the common detector-level post-processing selection rather than by production-time trigger/veto filtering. Third, the mission-time calculation is a rate-level fold, not a new particle transport at each trajectory time bin. Fourth, the custom Compton/field-of-view veto is a detector-level topology and kinematic consistency check, not yet a full Revan/Mimrec reconstruction, and the retained reconstruction-failure class is a baseline acceptance choice rather than a conservative sensitivity bound. Fifth, the event-level analysis has made the cut-flow, broad-window spectra, veto behavior, and activation selected-event decomposition traceable, but selected delayed state/metastable labels, the merged four-sampling per-event decomposition, post-veto energy-window staging, multi-pixel residual distributions, and veto-threshold stability scans are still not closed by the current outputs.

More generally, the tabulated 3σ flux thresholds are statistical selected-rate thresholds for the reference source, mass, and response model. They do not include a systematic-uncertainty envelope for atmospheric-

flux normalization, activation cross sections or physics-list choices, material and geometry tolerances, detector energy-response uncertainty, veto-threshold calibration, reconstruction-efficiency systematics, or background-estimation nuisance parameters. A publication-level flight sensitivity should replace the diagnostic $S/\sqrt{B}$ metric with a spatial-spectral likelihood, following standard profile-likelihood practice [Cowan et al.(2011)].

The upstream Laue-lens calculation in Section ?? should be read as a boundary calculation for an equal-volume/equal-mass active-Ge proxy, not as a finalized optics-background budget. It completes full-particle prompt and activation-production transport for the proxy, and the isolated Ge-proxy delayed component gives zero selected W_{511} events in the present transport. The prompt self-background contribution is still not part of the primary detector-background budget because it needs source separation or subtraction from the ordinary detector/cryostat prompt background, and explicit lens support hardware is not yet modeled.

Spatial analyses such as r_{90} and multi-annulus likelihood estimates are diagnostic cross-checks, but they are not yet a finalized nuisance-profile likelihood. The quoted sensitivity therefore remains the line-window selected-rate result, not a spatially optimized likelihood result.

Before this rate study can support a flight-performance statement, four validation blocks remain: a Re-van/Mimrec or equivalent reconstruction cross-check of the retained multi-hit subset, including residual distributions and threshold stability; optics self-background transport with explicit lens tiles and support hardware at final statistics; a final cryostat/DR geometry and detector-response update, including TES saturation and pile-up constraints; and a spatial-spectral likelihood that profiles background nuisance parameters. The event-level tables added here should be read as manuscript support evidence and audit entry points before those validations, not as a complete systematic-uncertainty envelope.

4.2 Conclusions

We have presented a detector-coupled background and sensitivity study for a balloon-borne pointed 511 keV TES Laue-lens telescope. The reference calculation couples a Ge(111) Laue optical response to a TES focal-plane mass model, prompt atmospheric transport, delayed activation, active-shield vetoing, topology/field-of-view selection, mission-time rate folding, and counting-significance evaluation. In the primary 510.58–511.42 keV unresolved-line window, the day-15 selected rates are $B \simeq 3.92 \times 10^{-2} \text{ s}^{-1}$ and $S \simeq 1.19 \times 10^{-3} \text{ s}^{-1}$ at $10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$; the mission-time fold gives $Z_{20d} \simeq 7.80$ and $F_{3\sigma}(20d) \simeq 3.85 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$. The result provides a detector-coupled selected-rate basis for continued evaluation of focused TES spectroscopy as a complementary 511 keV technique for compact-source and unresolved-central-excess searches, while identifying prompt atmospheric background suppression, delayed-activation systematic validation, and reconstruction validation as the main next validation tasks.

Data and software availability

Before publication, a versioned public repository or institutional archive should provide the geometry and source definitions, random seeds, analysis and figure-generation scripts, LaTeX source, and derived validation products needed to reproduce the tables and figures in this paper. This compile-ready recovery copy is not the original final manuscript.

Declarations

Funding To be completed before journal submission.

Competing interests To be completed before journal submission.

Author contributions To be completed before journal submission.

Ethics approval and consent to participate Not applicable.

References

[Prantzos et al.(2011)] N. Prantzos, C. Boehm, A.M. Bykov, R. Diehl, K. Ferriere, N. Guessoum, P. Jean, J. Knoedlseder, A. Marcowith, I.V. Moskalenko, A. Strong, G. Weidenspointner, The 511

keV emission from positron annihilation in the Galaxy, *Rev. Mod. Phys.* 83 (2011) 1001–1056, doi:10.1103/RevModPhys.83.1001.

- [Jean et al.(2006)] P. Jean, J. Knoedlseder, V. Lonjou, M. Allain, J.-P. Roques, G.K. Skinner, B.J. Teegarden, G. Vedrenne, P. von Ballmoos, B. Cordier, P. Caraveo, R. Diehl, P. Durouchoux, P. Leleux, R. Schanne, G. Weidenspointner, Spectral analysis of the Galactic e^+e^- annihilation emission, *Astron. Astrophys.* 445 (2006) 579–589, doi:10.1051/0004-6361:20053765.
- [Siegert et al.(2016)] T. Siegert, R. Diehl, G. Khachatryan, M.G.H. Krause, F. Guglielmetti, J. Greiner, A.W. Strong, X. Zhang, Gamma-ray spectroscopy of positron annihilation in the Milky Way, *Astron. Astrophys.* 586 (2016) A84, doi:10.1051/0004-6361/201527510.
- [Siegert et al.(2019)] T. Siegert, R. Diehl, G. Khachatryan, M.G.H. Krause, F. Guglielmetti, J. Greiner, A.W. Strong, X. Zhang, Constraints on positron annihilation kinematics in the inner Galaxy, *Astron. Astrophys.* 627 (2019) A126, doi:10.1051/0004-6361/201833856.
- [Yoneda et al.(2025)] H. Yoneda, T. Siegert, S. Mittal, Imaging the positron annihilation line with 20 years of INTEGRAL/SPI observations, arXiv:2509.01066, 2025.
- [Kierans et al.(2020)] C.A. Kierans, S.E. Boggs, A. Zoglauer, A.W. Lowell, C. Sleator, J. Beechert, T.J. Brandt, P. Jean, H. Lazar, J. Roberts, T. Siegert, J.A. Tomsick, P. von Ballmoos, Detection of the 511 keV Galactic positron annihilation line with COSI, *Astrophys. J.* 895 (2020) 44, doi:10.3847/1538-4357/ab89a9.
- [Siegert et al.(2020)] T. Siegert, S.E. Boggs, J.A. Tomsick, A.C. Zoglauer, C.A. Kierans, C.C. Sleator, J. Beechert, T.J. Brandt, P. Jean, H. Lazar, A.W. Lowell, J.M. Roberts, P. von Ballmoos, Imaging the 511 keV positron annihilation sky with COSI, *Astrophys. J.* 897 (2020) 45, doi:10.3847/1538-4357/ab9607.
- [Knoedlseder et al.(2005)] J. Knoedlseder, P. Jean, V. Lonjou, G. Weidenspointner, N. Guessoum, R. Diehl, W. Gillard, M.J. Harris, G.K. Skinner, P. von Ballmoos, G. Vedrenne, P. Caraveo, B. Cordier, V. Schoenfelder, B.J. Teegarden, The all-sky distribution of 511 keV electron-positron annihilation emission, *Astron. Astrophys.* 441 (2005) 513–532, doi:10.1051/0004-6361:20042063.
- [De Cesare(2011)] G. De Cesare, Searching for the 511 keV annihilation line from galactic compact objects with the IBIS gamma ray telescope, *Astron. Astrophys.* 531 (2011) A56, doi:10.1051/0004-6361/201116516.
- [Shirazi et al.(2023)] F. Shirazi, E. Gau, M.A. Hossen, D. Becker, D. Schmidt, D. Swetz, D. Bennett, D. Braun, F. Kislat, J. Gard, J. Mates, J. Weber, N.R. Cavero, S. Chun, L. Lisalda, A. West, B. Dev, F. Ferrer, R. Bose, J. Ullom, H. Krawczynski, The 511-CAM Mission: A pointed 511 keV gamma-ray telescope with a focal plane detector made of stacked transition edge sensor microcalorimeter arrays, *J. Astron. Telesc. Instrum. Syst.* 9 (2023) 024006, doi:10.1117/1.JATIS.9.2.024006.
- [Barriere et al.(2009)] N.M. Barriere, L. Natalucci, N. Abrosimov, P. von Ballmoos, P. Bastie, P. Courtois, M. Jentschel, J. Knoedlseder, J. Rousselle, P. Ubertini, Soft gamma-ray optics: new Laue lens design and performance estimates, arXiv:0910.0488, 2009.
- [Barriere et al.(2011)] N.M. Barriere, J.A. Tomsick, S.E. Boggs, A. Lowell, P. von Ballmoos, Developing a second generation Laue lens prototype: high reflectivity crystals and accurate assembly, arXiv:1111.6700, 2011.
- [Weidenspointner et al.(2006)] G. Weidenspointner, C.B. Wunderer, N. Barriere, A. Zoglauer, P. von Ballmoos, Monte Carlo study of detector concepts for the MAX Laue Lens gamma-ray telescope, arXiv:astro-ph/0603152, 2006.
- [Sanchez del Rio & Dejus(2011)] M. Sanchez del Rio, R.J. Dejus, XOP v2.4: recent developments of the x-ray optics software toolkit, *Proc. SPIE* 8141 (2011) 814115, doi:10.1117/12.893911.

- [Guan et al.(2023)] F. Guan, M. Asai, D.A. Bartkoski, M. Kleckner, Z. Harel, M. Salehpour, Adding the X-ray Bragg reflection physical process in crystal to the Geant4 Monte Carlo simulation toolkit, part I: reflection from a crystal slab, *Precision Radiation Oncology* 7(1) (2023) 59–66.
- [Reiazi et al.(2025)] R. Reiazi, D. Lee, D. Bartkoski, M. Kleckner, M. Salehpour, G4BraggReflection for accurate modeling of Bragg reflection in perfect and mosaic crystals, *J. Phys. D: Appl. Phys.* 58 (2025) 295102, doi:10.1088/1361-6463/aded1d.
- [Siegert et al.(2016)] T. Siegert, R. Diehl, J. Greiner, M.G.H. Krause, A.M. Beloborodov, M. Cadolle Bel, F. Guglielmetti, J. Rodriguez, A.W. Strong, X. Zhang, Positron annihilation signatures associated with the outburst of the microquasar V404 Cygni, *Nature* 531 (2016) 341–343, doi:10.1038/nature16978.
- [Gottardi & Nagayoshi(2021)] L. Gottardi, K. Nagayoshi, A review of X-ray microcalorimeters based on superconducting transition edge sensors for astrophysics and particle physics, *Applied Sciences* 11 (2021) 3793, doi:10.3390/app11093793.
- [Xie et al.(2026)] L. Xie, Y. Zhang, Z. Li, Z. Liu, S. Shu, J. Zhou, X. Li, H. Li, H. Gao, Y. Gu, X. Lu, Y. Zhao, C. Liu, Beyond one-thousandth energy resolution with an AlMn TES detector, arXiv:2602.11728, 2026.
- [Vavagiakis et al.(2017)] E.M. Vavagiakis, S.W. Henderson, K. Zheng, H.-M. Cho, N.F. Cothard, B. Dober, S.M. Duff, P.A. Gallardo, G. Hilton, J. Hubmayr, K.D. Irwin, B.J. Koopman, D. Li, F. Nati, M.D. Niemack, C.D. Reintsema, S. Simon, J.R. Stevens, A. Suzuki, B. Westbrook, Magnetic sensitivity of AlMn TESes and shielding considerations for next-generation CMB surveys, arXiv:1710.08456, 2017.
- [Hubbell & Seltzer] J.H. Hubbell, S.M. Seltzer, XCOM: Photon Cross Section Database, NIST Standard Reference Database 8 (XGAM), National Institute of Standards and Technology, doi:10.18434/T48G6X.
- [Zhang et al.(2022)] S. Zhang, J.-K. Xia, T. Sun, et al., Transition edge sensor based detector: from X-ray to gamma-ray, arXiv:2204.00010, 2022.
- [Gehrels(1985)] N. Gehrels, Instrumental background in balloon-borne gamma-ray spectrometers and techniques for its reduction, *Nucl. Instrum. Methods Phys. Res. A* 239 (1985) 324–349, doi:10.1016/0168-9002(85)90732-6.
- [Sato(2015)] T. Sato, Analytical model for estimating terrestrial cosmic ray fluxes nearly anytime and anywhere in the world: extension of PARMA/EXPACS, *PLoS ONE* 10 (2015) e0144679, doi:10.1371/journal.pone.0144679.
- [Sato(2016)] T. Sato, Analytical model for estimating the zenith angle dependence of terrestrial cosmic ray fluxes, *PLoS ONE* 11 (2016) e0160390, doi:10.1371/journal.pone.0160390.
- [Kondev et al.(2021)] F.G. Kondev, M. Wang, W.J. Huang, S. Naimi, G. Audi, The NUBASE2020 evaluation of nuclear physics properties, *Chinese Phys. C* 45 (2021) 030001, doi:10.1088/1674-1137/abddae.
- [Sturmer et al.(2003)] S.J. Sturmer, C.R. Shrader, G. Weidenspointner, B.J. Teegarden, D. Attie, B. Cordier, R. Diehl, C. Ferguson, P. Jean, A. von Kienlin, P. Paul, F. Sanchez, S. Schanne, P. Sizun, G. Skinner, C.B. Wunderer, Monte Carlo simulations and generation of the SPI response, *Astron. Astrophys.* 411 (2003) L81–L84, doi:10.1051/0004-6361:20031171.
- [Agostinelli et al.(2003)] S. Agostinelli, J. Allison, K. Amako, et al., Geant4—a simulation toolkit, *Nucl. Instrum. Methods Phys. Res. A* 506 (2003) 250–303, doi:10.1016/S0168-9002(03)01368-8.
- [Allison et al.(2016)] J. Allison, K. Amako, J. Apostolakis, et al., Recent developments in Geant4, *Nucl. Instrum. Methods Phys. Res. A* 835 (2016) 186–225, doi:10.1016/j.nima.2016.06.125.
- [Zoglauer et al.(2006)] A. Zoglauer, R. Andritschke, F. Schopper, MEGAlib: The Medium Energy Gamma-ray Astronomy Library, *New Astron. Rev.* 50 (2006) 629–632, doi:10.1016/j.newar.2006.06.049.

- [Caputo et al.(2017)] R. Caputo, F. Kislat, J. Racusin, on behalf of the AMEGO Team, AMEGO: Simulations of the instrument performance, Proc. 35th International Cosmic Ray Conference, PoS(ICRC2017) 783 (2018), doi:10.22323/1.301.0783.
- [Karwin et al.(2023)] C.M. Karwin, T. Siebert, J. Beechert, J.A. Tomsick, T.A. Porter, M. Negro, C. Kierans, M. Ajello, I. Martinez Castellanos, A. Shih, A. Zoglauer, S.E. Boggs, Probing the Galactic diffuse continuum emission with COSI, arXiv:2310.12206, 2023.
- [Gallego et al.(2025a)] S. Gallego, U. Oberlack, J. Lommler, C.M. Karwin, A. Zoglauer, P. Jean, P. von Ballmoos, C. Kierans, C.C. Sleator, J.A. Tomsick, S.E. Boggs, Bottom-up background simulations of the 2016 COSI balloon flight, arXiv:2503.02493, 2025.
- [Gallego et al.(2025b)] S. Gallego, U. Oberlack, J. Lommler, C.M. Karwin, F. Fenu, V. Fioretti, A. Zoglauer, et al., Pre-flight background estimates for COSI, arXiv:2510.25304, 2025.
- [Beechert et al.(2022)] J. Beechert, H. Lazar, S.E. Boggs, T.J. Brandt, Y.-C. Chang, C.-Y. Chu, H. Gulick, C. Kierans, A. Lowell, N. Pellegrini, J.M. Roberts, T. Siebert, C.C. Sleator, J.A. Tomsick, A. Zoglauer, Calibrations of the Compton Spectrometer and Imager, Nucl. Instrum. Methods Phys. Res. A 1031 (2022) 166510, doi:10.1016/j.nima.2022.166510.
- [Sleator et al.(2019)] C.C. Sleator, A. Zoglauer, A.W. Lowell, C.A. Kierans, S.E. Boggs, J.A. Tomsick, H. Lazar, J. Beechert, T.J. Brandt, P. von Ballmoos, Benchmarking simulations of the Compton Spectrometer and Imager with calibrations, Nucl. Instrum. Methods Phys. Res. A 946 (2019) 162643, doi:10.1016/j.nima.2019.162643.
- [Ciabattini et al.(2025)] A. Ciabattini, V. Fioretti, J.A. Tomsick, A. Zoglauer, P. Patel, L. Mitchell, A. Bulgarelli, P. Jean, G. Panebianco, N. Parmiggiani, C. Vignali, P. von Ballmoos, E. Wulf, Benchmarking of Geant4 simulations for the COSI anticoincidence system, Experimental Astronomy (2025), doi:10.1007/s10686-025-10019-7.
- [Cowan et al.(2011)] G. Cowan, K. Cranmer, E. Gross, O. Vitells, Asymptotic formulae for likelihood-based tests of new physics, Eur. Phys. J. C 71 (2011) 1554, doi:10.1140/epjc/s10052-011-1554-0.